

Ritabrata Chakraborty

[Github](#) | [Email](#) | [Google Scholar](#) | [Linkedin](#)

EDUCATION

Manipal University Jaipur

BTech in Computer Science | GPA : **9.37/10.0**

2022-Present

Top 1% of the School of Computer Science Engineering (SCSE)

Major Thesis: **Audio Description Generation for Movies under Linguistic and Cultural Shifts**

RESEARCH EXPERIENCE

Research Intern | [University of Manchester](#)

October 2025 - Present

- Supervisor : **Prof. Angelo Cangelosi**
- Worked on In-Context Mechanistic Binding for Vision Language Models, arising from Binding IDs literature for LLMs.
- Proposed **M2BIND**, investigating multilingual insight to vision-language concept binding through the lens of Whorfian Hypothesis and causality theory. [Paper under review.](#)
- *Ongoing Work*: Exploring spatial grounding in VLMs with **Visual Confusability**, proposing a new grounding framework for distinguishing within-category distractors. [Paper under preparation.](#)

Research Intern | [Indian Statistical Institute Kolkata](#), [University of Salford](#)

December 2023 – Present

- Supervisors: **Prof. Umapada Pal**, **Prof. Shivakumara Palaiahnakote**.
- Built a faster SAM inference pipeline for remote sensing and boosted guided segmentation via text prompting and **few-shot tuning**, delivering **30%** higher efficiency and **0.70** Dice on remote-sensing segmentation benchmarks.
- Worked towards utilization of contextual background cues for scene text recognition, providing comparable accuracy to SOTA methods at **10%** of FLOP cost. [Accepted as ICDAR 2025 Oral paper.](#)
- Developed **HATCHAR**, an early fusion model employing joint utilisation of online and offline features for handwritten text recognition, providing SOTA results for standard online-offline datasets. [Accepted as ACPR 2025 Oral paper.](#)
- Developed the **CDOD** benchmark for cross-dataset object detection under setting-specificity, leveraging CLIP-based open-label evaluation protocols. [Paper under review.](#)

Undergraduate Research Assistant | [International Institute of Information Technology, Hyderabad](#)

May 2025 – May 2026

- Supervisor: **Dr. Makarand Tapaswi**, Katha-AI group.
- Worked on improving and adapting audio descriptions for Hindi movies and audiences in collaboration with the Reliance AI Speech team.
- Introduced ANDHA-DHUN, the first ever dataset of ADs in Hindi.
- Studied AD translations through the lens of Skopos theory in film translations, highlighting the poor AD generation/translation capabilities of open-source and commercial language models. [Paper under review.](#)

Resident Researcher | [Lossfunk, Bangalore](#)

July 2025 - August 2025

- Collaborator: **Paras Chopra**, Project : **Bitvision**
- Benchmarked **10+ open-source VLMs** on QR code understanding tasks, investigating TTA strategies for decoding QRs.
- Developed a structure-aware CAM benchmark, CAMBENCH-QR supported by Lossfunk. [Paper accepted in Proceedings Track of CVPR Workshop \(XAI4CV\) 2026.](#)

Research Fellow | [International Institute of Information Technology, Hyderabad](#)

May 2024 - July 2024

- Supervisor: **Dr. C. V. Jawahar**
- Worked on Scene Text Recognition (STR) with a focus on qualitative research gaps for Indic languages, identified inefficiencies in **data processing**.
- Developed a batch-wise **data augmentation** pipeline, resulting in a **200%** reduction in **pre-processing time** for images.

Research Intern | [Purdue University](#)

Jan 2024 - May 2024

- Supervisor: **Prof. Vaneet Aggarwal**
- Reproduced results for POSNEGDM, a positive-negative trajectory based RL model for sepsis treatment.
- Experimented around incorporating **explainability** into the model, aiming to improve human trust to the medical community using Grad-CAM based explanations.

Undergraduate Research Assistant | [Manipal University Jaipur](#)

May 2023 – Oct 2024

- Supervisor: **Dr. Sandeep Chaurasia**
- Developed a "pair-therapist" interface for anonymously assessing the severity of cyberbullying and hazing cases reported by university students, adopted within MUJ's student counseling services.
- Prototyped **MANN-MITRA**, an ensemble framework for depression detection leveraging Q/A, gaze, and speech analysis.

PATENTS

Chakraborty R., Chaurasia S. (2023). "MANN-MITRA: A Depression Detection System Using Machine Learning." [\[Link\]](#)

Chakraborty R., Chaurasia S. (2024). "NAVYA: Navigational Audio for Visual Yield Assistance." [\[Link\]](#)

Paul A., **Chakraborty R.**, Chaurasia S. (2025). "A System for Geographically Contextual Crater Detection" [\[Link\]](#)

PUBLICATIONS

- Chakraborty R.**, Shivakumara P., Pal U., Liu C.-L. (2025). "A Lightweight Context-Driven Training-Free Network for Scene Text Segmentation and Recognition," **International Conference on Document Analysis and Recognition (ICDAR 2025)** [Oral] [Website] [Paper] [Code]
- Lodh A.*, **Chakraborty R.***, Palaiahnakote S., Pal U. (2025). "A Transformer Based Handwriting Recognition System Jointly Using Online and Offline Features." **Asian Conference on Pattern Recognition (ACPR 2025)** [Oral] [Paper] [Code]
- Chakraborty R.**, Chakraborty R., Rahimian A. K. (2025). "TruthLens: Training-Free Data Verification for Deepfake Images via VQA-style Probing," **ICML Workshop on Data in Generative Models 2025** [Paper]
- Chakraborty R.**, Chakraborty R., Dasgupta A., Chaurasia S. (2025). "Do We Need Large VLMs for Spotting Soccer Actions?" **IJCNLP-AAACL 2025 Student Research Workshop** [Oral] [Paper]
- Chakraborty R.**, Dasgupta A., Chaurasia S. (2025). "CAMBench-QR: A Structure-Aware Benchmark for Post-Hoc Explanations with QR Understanding." **Proceedings of CVPR Workshop (XAI4CV) 2026** [Paper] [Code]
- Chakraborty R.***, Mitra H.*, Palaiahnakote S., Pal U. (2026). "Towards Robust Cross-Dataset Object Detection Generalization under Domain Specificity" [Under review](#) [Paper] [Code]
- Chakraborty R.**, Chakraborty R., Palaiahnakote S., Pal U., Cangelosi A. (2026). "Vision-Language Models are Fragile Multilingual Associators" [Under review](#)
- Chakraborty R.**, Kala D., Gupta N., Ganji S., Balakrishna P., Tapaswi M. (2026). "ANDHA-DHUN: A First Look at Audio Descriptions in Hindi" [Under review](#)

PROJECTS

- Making Paris look like Paris : Visual Concept Profiling for Locations** | *IIIT Hyderabad* *Location-Aware Vision*
- An independent study of visual concept discovery in real images for building a profile of cities.
 - Evaluated city concept profiles using **JSD²** and **Sinkhorn OT distance**, with residual analysis on **3k real images** scraped from **Flickr** and **Wikimedia** across **10 cities**, and on **SDXL-generated images** for the same cities.
- TruthLens: A Training-Free Paradigm for DeepFake Detection** | *Manipal University Jaipur* *Computer Vision, AI Ethics*
- Designed **TruthLens**, a **training-free deepfake detection framework** leveraging Large Vision-Language Models (LVLMs) and LLMs. [Workshop paper in ICML DIGBUGS 2025.](#)
 - Transformed deepfake detection into a **Visual Question Answering (VQA) task**, improving interpretability.
 - Achieved state-of-the-art detection accuracy while providing **explainable insights** for detecting synthetic media.

TEACHING EXPERIENCE

- Undergraduate Teaching Assistant** | *Manipal University Jaipur* *May 2025 – Dec 2025*
- Serving as **Lead Teaching Assistant** for **CS3203 — Data Science and Machine Learning** and **IT3241 — Computer Vision**.
 - Conducting **recitations, grading assignments, and mentoring students** on classical computer vision and statistical machine learning concepts.
 - Assisting with **course material development**, ensuring alignment with latest research directions.

TECHNICAL SKILLS AND INTERESTS

Languages : Python, C/C++, Java, MySQL
Frameworks : Pytorch, Tensorflow, Tensorflow Extended (TFX)
DevOps and API Tools : Github,Weights and Biases, Git
Organizations : Founder @ [Vision Language Research \(VLR\) MUJ](#), Projects and Research Head @ [ACM SIGAI MUJ](#)

HONORS AND AWARDS

Dean's List (x8) | *Manipal University Jaipur* | Awarded **Highest GPA** *2022-2026*
Summer Research Fellowship Program 2024 | *IAS-NASI-INSa* | Awarded to exclusively **200 students nationwide** *May 2024*
Student Excellence Award (x6) | *Manipal University Jaipur* | Recognizing **extracurricular efforts** *June 2024*

SERVICES AND VOLUNTEERING

Reviewer | *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)* *January 2026*
Reviewer | *IEEE Transactions on Image Processing (TIP)* *August 2025*
Conference Student Host | *Bharat Blockchain Yatra* *Feb 2024*